# Walchand Institute of Technology, Solapur
(An Autonomous Institute)
Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur
Choice Based Credit System (CBCS)
Structure and Syllabus
for
Multidisciplinary Minor Program in
Civil Engineering
offered by
Department of Civil Engineering
for Batch Admitted in

2024-2025



HEAD,
Department of Civil Engineering
Walchand Institute of Technology,
SOLAPUR-413006.

WALCHAND INSTITUTE OF TECHNOLOGY
WIT SOLAPUR (Autonomous)

Dr. Mrs. M. A. Nirgude
Dean Academics

| Multidisciplinary Minor Program in Civil Engineering | |
|---|---|
| **Legends Used** | |
| L | Lecture Hours / week |
| T | Tutorial Hours / week |
| P | Practical Hours / week |
| FA | Formative Assessment |
| SA | Summative Assessment |
| ESE | End Semester Examination |
| ISE | In Semester Evaluation |
| ICA | Internal Continuous Assessment |
| POE | Practical and Oral Exam |
| OE | Oral Exam |
| MOOC | Massive Open Online Course |
| HSS | Humanity and Social Science |
| NPTEL | National Programme on Technology Enhanced Learning |
| F.Y. | First Year |
| S.Y. | Second Year |
| T.Y. | Third Year |
| B. Tech. | Bachelor of Technology |

# Multidisciplinary Minor Program in Civil Engineering

| **Course Code Format** | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **2** | **1** | **C** | **E** | **U/P** | **2** | **C** | **C** | **1** | **T/L** |
| Year of Syllabus revision | | Program Code | | U-Under Graduate P-Post Graduate | Semester No./ Year1/2/3/…8 | CourseType | | Course Serial No1-9 | T-Theory, L-Lab session A-Tutorial P-Programming / Design |

| | |
|---|---|
| **Program Code** | |
| CE | Civil Engineering |
| **Course Type** | |
| BS | Basic Science |
| ES | Engineering Science |
| HU | Humanities & Social Science |
| MC | Mandatory Course |
| CC | Programme Core Compulsory Course |
| SN* | Self-Learning (**N*** *indicates the serial number of electives offered in the respective category)* |
| EN* | Programme Core Elective Course (**N*** *indicates the serial number of electives offered in the respective category)* |
| SK | Skill-Based Course |
| SM | Seminar |
| MP | Mini project |
| PR | Project |
| IN | Internship |
| ON* | Open Elective (**N*** *indicates the serial number of electives offered in the respective category)* |
| MD | Multidisciplinary Minor |
| EM | Entrepreneurship/Economics/Management Courses |
| FP | Community Engagement Project / Field Project |

| | |
|---|---|
| AE | Ability Enhancement Course |
| VE | Value Education Course |
| IK | Indian Knowledge System |
| VS | Vocational & Skill Enhancement Course |
| RM | Research Methodology |
| HN | Honors' Degree Course |
| HR | Honors Research |

| **Sample Course Code** | |
|---|---|
| 24CEU3MD6T | Multidisciplinary Minor I |

# Civil Engineering

## Multidisciplinary Minor Program in Civil Engineering

| Semester | Course Code | Name of Course | Engagement Hours | | | Credits | SA | | FA | | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | L | T | P | | Theory | OE/ POE | ISE | ICA | |
| III | 24CEU3MD6T | Smart Buildings | 2 | - | - | 2 | 60 | - | 40 | - | 100 |
| IV | 24CEU4MD6T | Geoinformatics | 1 | - | - | 1 | - | - | 50 | - | 50 |
| V | 24CEU5MD6T | Environmental Impact Assessment | 3 | - | - | 3 | 60 | - | 40 | - | 100 |
| VI | 24CEU6MD6T | Infrastructural Systems | 2 | - | - | 2 | 60 | - | 40 | - | 100 |
| VII | 24CEU7MD6T | Disaster preparedness and planning | 3 | - | - | 3 | 60 | - | 40 | - | 100 |
| VII | 24CEU7MD6A | Disaster preparedness and planning | - | 1 | - | 1 | - | --- | | **25** | 25 |
| | | **Subtotal** | **11** | **1** | | **12** | **240** | **-** | **210** | | **475** |
| **Laboratory Courses** | | | | | | | | | | | |
| IV | 24CUE4MD6L | Geoinformatics Lab | - | - | 2 | 1 | - | - | | 25 | 25 |
| VI | 24CEU6MD6L | Infrastructural Systems Lab | - | - | 2 | 1 | - | - | | 25 | 25 |
| | | **Subtotal** | **11** | **-** | **4** | **2** | **-** | **-** | | **50** | **50** |
| | | **Grand Total** | **11** | **1** | **4** | **14** | **240** | **-** | **210** | **75** | **525** |

| WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR<br>(An Autonomous Institute)<br>Second Year B.Tech. (Multidisciplinary Minor in Civil Engineering), Semester-III | | | |
|---|---|---|---|
| **24CEU3MD6T: Multidisciplinary Minor I-Smart Buildings** | | | |
| **Teaching Scheme** | | **Examination Scheme** | |
| **Lectures** | 2 Hours/week | **ESE** | 60 Marks |
| **Credits** | 2 | **ISE** | 40 Marks |

**Introduction:**

This course provides a thorough introduction to Smart Building Technology and its concepts related to building Management Systems for the development of intelligent and smart buildings.

**Course Prerequisite:**

Basic understanding of various units of building units and their functions

**Course Objectives:**

1. To enhance the knowledge of Smart Building Technologies
2. To know about the concept of Building Automation
3. To evaluate the functioning of Smart Buildings.

**Course Outcomes:**

After completing the course, students will be able to,

1. Recognize the system of Smart Technology
2. Identify the technology involved in the Integrated Building Management System (IBMS) in Smart buildings
3. Illustrate the philosophy of building Management systems and IoT.
4. Comprehend the technology available in Heating, Ventilation, and Air Conditioning (HVAC) and Lighting control systems.
5. Explain technology available in fire safety and security systems for intelligent buildings.

| | | |
|---|---|---|
| **Unit – I** | **Introduction to Smart Buildings** | **2 Hours** |
| Definition and Overview, History and evolution, Benefits and challenges | | |
| **Unit – II** | **Integrated Building Management System (IBMS) in Smart Buildings** | **5 Hours** |
| ▪ Introduction to Integrated Building Management System (IBMS)Smart buildings.<br>▪ Components of Smart Building IBMS. | | |
| **Unit – III** | **Building Management Systems (BMS) and** | **5 Hours** |

| | | |
|---|---|---|
| | **IoT** | |
| ▪ Building Management Systems (BMS): Components of BMS, Functions and Architecture.<br>▪ Internet of Things (IoT) in Smart Buildings: IoT fundamentals: IoT devices and sensors, Communication protocols. | | |
| **Unit – IV** | **Heating, Ventilation, and Air Conditioning (HVAC) and Lighting Controls Systems:** | **6 Hours** |
| ▪ Human Comfort: Air Quality-Thermal comfort in buildings-Ventilation-Classification of HVAC - DDC applications - Control Panel: HVAC Control Panel, Networks, BAC Net, Modbus, LON-Electrical Installation Power Transmission -Smart Lighting Systems | | |
| **Unit – V** | **Fire Protection Systems:** | **6 Hours** |
| ▪ Introduction: Causes and Stages of Fire, Fire Service Installation, Fire Alarm. Principles of Operation, Fire Sensors: Smoke detectors and their types, Fire control panels, Fire and Life safety, and Fire Management in buildings. | | |
| **Unit – VI** | **Security Systems** | **6 Hours** |
| ▪ Introduction to Security Systems: Concepts, Perimeter Intrusion, Security Design, Access control system, RFID-enabled access control, Computer system access control: CCTV, Components of CCTV system, Central alarm systems, and Structural health monitoring systems. | | |
| **INTERNAL CONTINUOUS ASSESSMENT (ICA)**<br>Internal Continuous Assessment (ICA) shall consist of a minimum of five assignments based on the entire curriculum and which will be assessed by the concerned faculty. | | |
| **Text Books** | | |
| 1. Shengwei Wang, “Intelligent Buildings and Building Automation”, Spon Press, London, 2009.<br>2. Derek Clements and Croome, “Intelligent Buildings: An Introduction”, Routledge, 2013, Civil Engineering. | | |
| **Reference Books** | | |
| 1. Derek Clements and Croome, “Intelligent Buildings: An Introduction”, Routledge, 2013<br>2. James Sinopoli “ Advanced Technology for Smart Buildings”, Artech House, 2016<br>3. James.M.Sinopoli, Smart Buildings Systems for Architects, Owners and Builders Publishers:” Butter worth Heinemann, 2009.<br>4. James Kachadorian “ Passive Solar House: the complete Guide to Heating and Cooling Your Home “ Chelsa Green Publishing: Revised and expanded Second edition, 2006 Ron Bakker, “Smart Buildings: Technology and the Design of the Built Environment”, RIBA Publishing, 2020. | | |
| **e-Resources** | | |
| 1. NPTEL (India): Introduction to Smart Grid-https://nptel.ac.in<br>2. edX: Smart Buildings and the Internet of Things (offered by TU Delft)-https://www.edx.org | | |

| WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR<br>(An Autonomous Institute)<br>Second Year B.Tech. (Multidisciplinary Minor in Civil Engineering), Semester-IV |
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## 24CEU4MD6T: Multidisciplinary Minor II-Geoinformatics

| **Teaching Scheme** | | **Examination Scheme** | |
|---|---|---|---|
| **Lectures** | 1 Hour/week | **ISE** | 50 Marks |
| **Practicals** | 2 Hours/week | **ICA** | 25 Marks |
| **Credits** | 2 | | |

**Introduction:**

This course covers the fundamentals of remote sensing, GIS, GPS, and geoinformatics applications essential for spatial data analysis. Topics include remote sensing principles, GIS components and data structures, GPS technology and geo-positioning techniques, and geoinformatics applications in natural resource management, environmental studies, and urban planning, emphasizing practical applications in engineering and environmental practice.

**Course Prerequisite:**

Basic knowledge of Physics, Geography, and Earth sciences

**Course Objectives:**

1. Explain remote sensing history, EMR interaction, platforms, sensors, and resolution types for analyzing satellite imagery data.
2. Understand GIS components, spatial and attribute data characteristics, raster vs. vector structures, and analyze satellite data.
3. Understand GPS history, segments, geo-positioning techniques, advantages, limitations, and practical applications in various fields.
4. Utilize geoinformatics for managing natural resources, environment, disasters, utilities, urban studies, military, navigation, and agriculture problems.

**Course Outcomes:**

After completing the course, students will be able to

1. Expalin remote sensing history, concepts, EMR interaction, platforms, sensors, and types of resolution in satellite imagery analysis.
2. Analyze and apply knowledge of GIS components, spatial and attribute data, raster vs. vector structures, and process satellite data in GIS.
3. Explain GPS history, advantages, limitations, segments, geo-positioning techniques, and practical applications across various fields.
4. Proficient to apply geoinformatics to manage natural resources, environment, disasters, utilities, urban areas, military, navigation, and agriculture, solving related problems.

| Unit – I | Introduction to Remote Sensing | 4 Hours |
|---|---|---|
| Remote sensing history, concept and principles, Interaction of EMR with atmosphere and Earth's surface, Remote sensing platforms and sensors, Concept of Resolution-spatial, spectral, temporal and radiometric, Satellite and their characteristics- geostationary and sun-synchronous | | |
| **Unit – II** | **Fundamentals of Geographic Information System:** | **4 Hours** |
| Components of GIS, Data used in GIS, characteristics of Spatial Data – sources of spatial and attribute data, data structure - raster and vector, Processing of various satellite data. | | |
| **Unit – III** | **Global Positioning System and its Applications** | **4 Hours** |
| History of GPS, Advantages and limitations of GPS, Segments of GPS - Control segment, Space segment, User segment, Geo positioning - Point positioning, Relative Positioning, Static Positioning, Kinematics Positioning, Uses of GPS. | | |
| **Unit – IV** | **Application of Geoinformatics:** | **3 Hours** |
| Application of Geoinformatics in Natural Resources Management, Environmental Studies, Disaster Management, Utilities Management, Urban Studies, Military Applications – Navigation - Location Based Services and Agriculture. | | |

**INTERNALCONTINUOUSASSESSMENT(ICA)**

Internal Continuous Assessment (ICA) shall consist of a minimum of five assignments based on the entire curriculum and which will be assessed by the concerned faculty.

## Text Books

1. Shengwei Wang, "Intelligent Buildings and Building Automation", Spon Press, London, 2009.
2. Derek Clements and Croome, "Intelligent Buildings: An Introduction", Routledge, 2013, Civil Engineering.

## Reference Books

1. Derek Clements and Croome, "Intelligent Buildings: An Introduction", Routledge, 2013
2. James Sinopoli, " Advanced Technology for Smart Buildings", Artech House, 2016
3. James.M.Sinopoli, Smart Buildings Systems for Architects, Owners and Builders, Publishers:" Butterworth Heinemann, 2009.
4. James Kachadorian, " Passive Solar House: The Complete Guide to Heating and Cooling Your Home, "Chelsea Green Publishing: Revised and expanded Second edition, 2006. Ron Bakker, "Smart Buildings: Technology and the Design of the Built Environment", RIBA Publishing, 2020.
5. A Course in Civil Engineering Drawing, Sikka, S.K., Kataria & Sons
6. Engineering Drawing, Dhanarajay A. Jolhe, Tata McGraw-Hill

**e-Resources**

1. NPTEL-Geoinformatics for Natural Resource Management by IIT Roorkee-nptel.ac.in
2. Coursera-GIS, Mapping, and Spatial Analysis Specialization (UC Davis)-coursera.org

| WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR<br>(An Autonomous Institute)<br>Third Year B.Tech. (Multidisciplinary Minor in Civil Engineering), Semester-V | | | |
|---|---|---|---|
| **24CEU5MD6T: Multidisciplinary Minor III- Environmental Impact Assessment** | | | |
| **Teaching Scheme** | | **Examination Scheme** | |
| **Lectures** | 3 Hours/week | **ESE** | 60 Marks |
| **Practical** | - | **ISE** | 40 Marks |
| **Credits** | 3 | **ICA** | - |

**Introduction:**

This course covers the principles, methodologies, and processes of Environmental Impact Assessment (EIA), focusing on evaluating environmental impacts, mitigation strategies, and regulatory compliance in real-world development projects.

**Course Prerequisite:**

Basics of Environmental Engineering

**Course Objectives:**

1. To introduce students to the concept and importance of Environmental Impact Assessment in sustainable development.
2. To provide knowledge on the methodologies used for EIA and the preparation of Environmental Impact Statements (EIS).
3. To familiarize students with the legal and regulatory framework for EIA in various countries.
4. To enable students to analyze the environmental impacts of different industrial, infrastructural, and developmental projects.

**Course Outcomes:**

After completing the course, students will be able to
1. Describe the principles, objectives, and importance of EIA in environmental management.
2. Identify key components of an Environmental Impact Assessment and the methodology used to conduct assessments.
3. Evaluate the environmental impacts of various development projects.
4. Prepare Environmental Impact Statements (EIS) and suggest mitigation measures for identified impacts.

| | | |
|---|---|---|
| **Unit – I** | **Introduction to EIA** | **6 Hours** |

Definition and Scope of EIA, Importance of EIA in sustainable development, Evolution and historical perspective of EIA, Types of EIA: Project-specific and Strategic Environmental Assessment (SEA),

| | | |
|---|---|---|
| EIA Process and Methodology | | |
| **Unit – II** | **EIA Methodologies** | **6 Hours** |
| Screening, Scoping, and Public Participation, Baseline Data Collection and Impact Prediction, Tools and Techniques for Impact Assessment: Environmental Indicators, Mathematical Modelling for Impact Prediction, GIS and Remote Sensing in EIA, Environmental Impact Statement (EIS) preparation | | |
| **Unit – III** | **Environmental Impact Evaluation** | **8 Hours** |
| Identifying significant impacts, Impact classification: Positive vs. Negative, Direct vs. Indirect, Social, Economic, and Cultural Impact Assessment, Quantitative vs. Qualitative Assessment, Impact Significance Criteria, Risk Assessment in EIA | | |
| **Unit – IV** | **EIA for Different Sectors** | **8 Hours** |
| Industrial Projects: Manufacturing, Mining, etc., Infrastructure Projects: Highways, Dams, Airports, etc., Urban Development and Land Use Change, Environmental Impacts of Agriculture, Forestry, and Fisheries, Case Studies of sector-specific EIA | | |
| **Unit – V** | **Legal and Regulatory Framework** | **6 Hours** |
| National and International EIA Regulations, Role of Government Agencies and NGOs in EIA, Environmental Laws and Policies, Environmental Clearance Process, EIA in Developing Countries and Challenges | | |
| **Unit – VI** | **Mitigation and Monitoring** | **6 Hours** |
| Mitigation Strategies for Environmental Impacts, Environmental Management Plan (EMP), Post-EIA Monitoring and Evaluation, Case Study of EIA in practice: Successful EIA projects and failures | | |
| The ICA includes at least 6 assignments, where students are required to develop Environmental Impact Assessment (EIA) reports and related applications for real-world development projects | | |
| **Text Books** | | |
| 1. Morrison-Saunders, A., Advanced Introduction to Environmental Impact Assessment, Edward Elgar Publishing, 2nd Edition, 2023.<br>2. Islam, K. M. B., & Nomani, Z. M. (Eds.), Environment Impact Assessment: Precept & Practice, CRC Press, 1st Edition, 2022.<br>3. Bhateria, R., Sharma, M., Singh, R., & Kumar, S., Environmental Impact Assessment: A Journey to Sustainable Development, Springer, 1st Edition, 2024. | | |

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| 4. Rosales, J., Environmental Impact Assessment, Delve Publishing, 1st Edition, 2021. |
| **Reference Books** |
| 1. Fonseca, A. (Ed.), Handbook of Environmental Impact Assessment, Edward Elgar Publishing, 1st Edition, 2022.<br>2. Hanna, K. (Ed.), Routledge Handbook of Environmental Impact Assessment, Routledge, 1st Edition, 2022.<br>3. Hosetti, B. B., & Kumar, A., Environmental Impact Assessment and Management, Daya Publishing House, 1st Edition, 2024.<br>4. Carroll, B., Environmental Impact Assessment Handbook: A Practical Guide for Planners, Developers and Communities, Emerald Publishing Limited, 3rd Edition, 2019. |
| **e-Resources** |
| 1. NPTEL Course: Environmental Impact Assessment<br>2. Coursera Course: Introduction to Environmental Impact Assessment |

**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR**

**(An Autonomous Institute)**

**Third Year B.Tech. (Multidisciplinary Minor in Civil Engineering), Semester-VI**

## 24CEU6MD6T: Multidisciplinary Minor IV-Infrastructural Systems

| Teaching Scheme | | Examination Scheme | |
|---|---|---|---|
| **Lectures** | 2 Hours/week | **ESE** | 60 Marks |
| **Practical** | 2 Hours/week | **ISE** | 40 Marks |
| **Credits** | 3 | **ICA** | 25 Marks |

**Introduction:**

This course provides a comprehensive understanding of the planning, design, construction, and management of critical infrastructure systems, including transportation, water supply, energy, telecommunications, and waste management.

**Course Prerequisite:**

Basic understanding of civil engineering principles, Environmental science, and Surveying and Geomatics

**Course Objectives:**

1. Understand the key components and interdependencies of infrastructural systems.
2. To examine the components and technologies involved in transportation, water, and wastewater infrastructure.
3. To explore the concept, objectives, and development frameworks of smart cities.
4. To analyse the planning, integration, and sustainability of smart city infrastructure, including energy, housing, and safety systems.
5. To evaluate and apply digital and intelligent tools such as GIS, GPS, and ITS for managing urban mobility and infrastructure.

**Course Outcomes:**

After completing the course, students will be able to

1. Describe various types of infrastructure systems and their relevance to socio-economic development.
2. Differentiate and illustrate the structural and functional aspects of transportation and water/wastewater systems.
3. Describe the concept and evolution of smart cities, along with the associated challenges and goals.
4. Analyse the integration of sustainable practices into smart city infrastructure, such as energy systems, green buildings, and disaster management.
5. Apply smart infrastructure tools (GIS, ITS, GPS) to develop and manage transportation and mobility services.

| | | |
|---|---|---|
| **Unit – I** | **Introduction to Infrastructure Systems** | **4 Hours** |
| Definition and importance of infrastructure, Classification of infrastructure (transportation, water, energy, etc.), Overview of urban and rural infrastructure, Stakeholders in infrastructure development, and Infrastructure planning process and regulatory framework | | |
| **Unit – II** | **Transportation Infrastructure** | **4 Hours** |
| Roadways, railways, airports, and ports, Urban transportation systems (metro, BRT, etc.), Intelligent Transportation Systems (ITS), and Case studies of transportation projects | | |
| **Unit – III** | **Water and Wastewater Infrastructure** | **4 Hours** |
| Water supply systems (sources, treatment, distribution), Wastewater management (collection, treatment, disposal), Stormwater drainage and flood control, and Integration with urban planning and sustainability aspects | | |
| **Unit – IV** | **Fundamentals of Smart City and Infrastructure** | **6 Hours** |
| Introduction of Smart City, Concept of smart city, Objective for smart cities, History of Smart City World and India. Need to develop a smart city, Challenges of managing infrastructure in India and the world, various types of infrastructure systems, Infrastructure needs assessment | | |
| **Unit – V** | **Planning and development of Smart city Infrastructure** | **6 Hours** |
| Energy and ecology, solar energy for smart city, Housing, sustainable green building, safety, security, disaster management, economy, cyber security, Project management | | |
| **Unit – VI** | **Tools for Smart City Infrastructure** | **6 Hour** |
| Intelligent Transport Systems, Smart Vehicles and Fuels, GIS, GPS, Navigation Systems, Traffic Safety Management, Mobility Services, E-ticketing | | |
| ICA consists of a minimum of 06 assignments based on the above syllabus | | |
| **Text Books** | | |
| 1. Grigg, N. S. Infrastructure Engineering and Management, McGraw Hill Education, Latest Edition. (Comprehensive coverage of infrastructure systems with planning and management aspects)<br>2. Thagesen, B. Highway and Traffic Engineering, Thomas Telford Publishing. (For transportation infrastructure details)<br>3. Moser, D. A., & Markus, M. D. Engineering Economic Analysis of Water Supply Systems, American Water Works Association. (Water infrastructure economics and planning).<br>4. Infrastructure: A Guide to the Industrial Landscape – Brian Hayes<br>5. Civil Infrastructure Systems: Intelligent Renewal – Hojjat Adeli<br>6. Sustainable Infrastructure: Principles into Practice – Charles A. Reed & Brian Robert | | |
| **Reference Books** | | |

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|---|
| 1. Sullivan, A. M., & Sheffrin, S. M. *Economics: Principles in Action*, Pearson Education.<br>2. Hansen, W. G. *Urban Planning and Infrastructure*, MIT Press.<br>**3.** Patra, S. C. *Municipal Engineering: Principles and Practice*, Narosa Publishing. |
| **e-Resources** |
| 1. www.worldbank.org<br>2. www.asce.org<br>3. https://elibrary.worldbank.org<br>**4.** https://smartcities.gov.in |